

BAO-NGOC PHAN

louisa.pbn@gmail.com | [GitHub](#) | [LinkedIn](#)

EDUCATION

National Tsing Hua University

Bachelor of Science in Materials Science and Engineering.

Hsinchu, Taiwan

2026 – 2030 (*expected*)

Le Hong Phong High School for the Gifted

Specialization: Chemistry

Ninh Binh, Vietnam

2023 – 2026

GPA: 9.5/10.0 · AP Chemistry 5 · AP Calculus AB & BC 5

- Achieved the only perfect 10 in specialized Chemistry in the 30-year history of the High School Entrance Exam.

RESEARCH EXPERIENCE

Lumiere Research

Mar 2026 – May 2026

Project: Evaluating surface functionalization strategies in carbon nanotube-based drug delivery for cancer therapy: A review

Advisor: Dr. Simran Kerai

- Conducting a systematic literature review on carbon nanomaterials in drug delivery, biosensing, and biomedical diagnostics; analyzing structure–property relationships and functionalization strategies to improve nanomaterial biocompatibility.

Center for Computational Science @HNUE

Research Assistant

Hanoi, Vietnam

Sep 2025 – present

Project: Tuning the Emission and Circularly Luminescence Dissymmetry of CzBN-Based Heterohelicenes,

(under review – 3rd author)

Supervisor: Professor Hue Minh Thi Nguyen

- Performed density functional theory (DFT) calculations using Gaussian and VASP for research projects; prepare figures, tables, and visualizations for reports, academic papers, and presentations; perform statistical analyses and interpret results to support research hypotheses.

SKILLS

Technical Skills: Computational Chemistry (Gaussian 09, VASP (beginner), Chemcraft); Programming (JavaScript, Python, C++); Development Tools (GitHub, VS Code); Graphic Design (Figma).

Language: Vietnamese (native); English (IELTS 7.0); Mandarin (TOCFL A1).

TEACHING ASSISTANT

HChemO

Teaching Assistant

Hanoi, Vietnam

July 2025 – present

- Developed supplemental study materials and practice problems to reinforce learning; supported teachers in attendance tracking, grading, and answering students' questions; provided targeted tutoring on AP Chemistry topics, including kinetics, equilibrium, and electrochemistry.

ADDITIONAL AWARDS/HONORS

- National Tsing Hua University International Student Scholarship
- Bronze Medal, Canadian Chemistry Contest (CCC) (2025)
- Gold Medal, C3L6 by University of Cambridge (2025)
- High Distinction, International Chemistry Quiz (ICQ) (2025)
- Bronze Medal, Hanoi University of Science Olympiad (HSGSO) (2024)
- Third Prize, Vietnam National University Olympiad (2024)
- Merit Award, Vietnam National Science – Chemistry Olympiad (2024)